

THE SUN EMITS MORE THAN JUST LIGHT

Sunlight contains ultraviolet (UV) radiation, a form of electromagnetic radiation that has a shorter wavelength and higher energy than visible light.

- UVA** Longest-wavelength, lowest-energy UV. It penetrates deep into the skin reaching the middle layers.
- UVB** Carries more energy. It's the primary cause of sunburn and plays the central role in skin cancer development.
- UVC** Highest-energy type and the most dangerous to living cells. It is mostly blocked by the atmosphere.

so, how do we protect our skin?

Sunscreens contain compounds that help reduce the amount of UV radiation reaching the skin.

AVOBENZONE

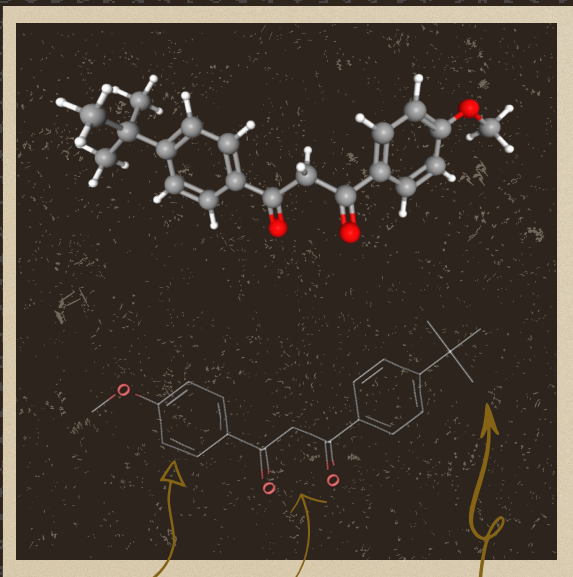
A topical, broad range UV protector and blocks UVA I, UVA II, and UVB wavelengths, thereby limiting the impact of UV rays on skin.

IUPAC name: 1-(4-tert-butylphenyl)-3-(4-methoxyphenyl)propane-1,3-dione

molecular formula: C₂₀H₂₂O₃

molar mass: 310.4 g/mol

Avobenzone is a member of dihydrochalcones and is dibenzoyl methane derivative.



Ketone (C=O)
a carbonyl group

Ether (C-O-C)
a methoxy group

Aromatic Rings
two benzene rings

Major Functional Groups

PHYSICAL AND CHEMICAL PROPERTIES

- a solid in its pure form and in room temperature
- crystals from methanol
- off-white to yellowish, crystalline powder
- weak odor, slightly aromatic
- 83.5 °C melting point
- nonpolar
- solubility at 20 °C: glycerin, 0.1% w/w, ethanol and isopropanol, 2% w/w
- water-insoluble, soluble in isopropanol, decyl oleate, capric/caprylic triglyceride, castor oil

Notable Structural Feature

Extended conjugated π -electron system

The structure contains alternating/conjugated π bonds across the aromatic rings and carbonyl-containing portion. This allows electrons to interact with and absorb UV radiation, particularly in the UVA region.

Sunscreen contains active ingredients called **UV filters**, which can absorb, scatter, or reflect UV radiation.

Avobenzone is particularly important for **UVA protection**. It absorbs strongly in the UVA region, approximately 320-400 nm.

BUT HOW DOES IT WORK?

UVA reaches the sunscreen

The conjugated π -electron system in the abenzone molecule absorbs energy from incoming UVA radiation.

The conjugated π -electron system in the abenzone molecule absorbs energy from incoming UVA radiation.

The molecule becomes electronically excited

The energy is released allowing absorbed energy to dissipate rather than allowing that same UV photon to reach the skin.

Because the radiation has been absorbed by the UV filter, the amount of UVA penetrating the skin is reduced.

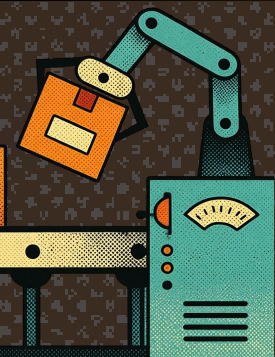
OTHER USES

- Fragrance and cosmetic formulations** for protection from light-induced degradation.
- Pharmaceutical research** as a model compound for developing methods to improve the photostability of light-sensitive compounds.
- Researchers particularly study Avobenzone as a model for developing methods that improve the stability of light-sensitive compounds.

RISKS & HAZARDZ

- May cause long-term harmful effects to aquatic life. Keep out of waterways.
- May cause skin irritation; rare photocontact reactions have been reported.
- Avoid dust formation and inhalation. Use gloves and appropriate eye protection.
- Fire can produce carbon oxides. Prevent environmental release and dispose of chemical waste properly.

FROM MOLECULE TO SUNSCREEN

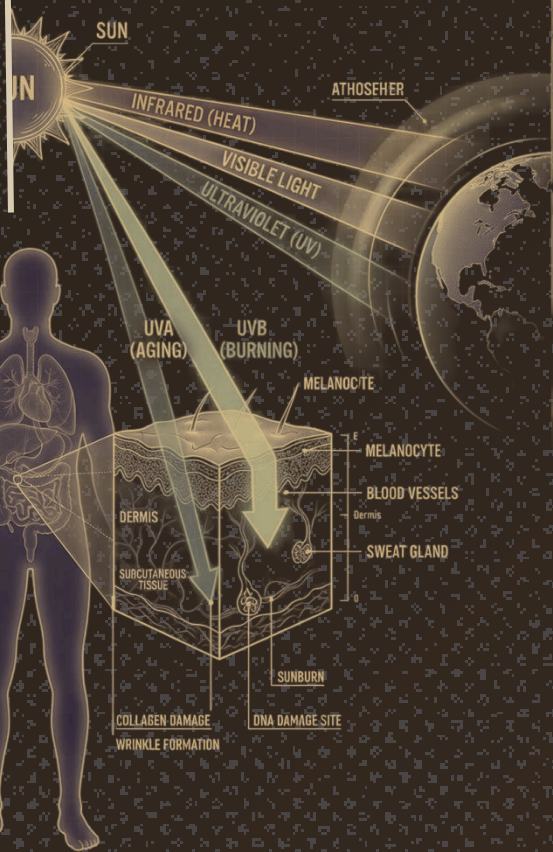


Raw Avobenzone
↓
Dissolution / mixing
↓
Formulation
↓
Emulsification
↓
Quality control
↓
Finished sunscreen

Avobenzone must be incorporated into a stable sunscreen formulation with appropriate solvents, oils, emulsifiers, and stabilizers. **Chemical engineers** can help optimize mixing, heat transfer, mass transfer, formulation stability, and product quality during manufacturing.

REFERENCES:

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PubChem. (n.d.). Avobenzone. PubChem. <https://pubchem.ncbi.nlm.nih.gov/compound/Avobenzone#section=Mechanism-of-ActionH22O3%20|%20CID%2051040%20-%20PubChem>



THE CHEMISTRY BEHIND SUN PROTECTION

